DE–3842

Sub. Code
11

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. List down six elements of a computer and Communications system. Also explain in detail. (20)

2. (a) Give a note on connectivity and interactivity of technology. (10)

   (b) What are the different types of application software? Explain. (10)

3. Write notes on the following in detail with creative examples :

   (a) word processing (10)

   (b) spreadsheets. (10)

4. (a) Discuss the concept of voice and video Communications of the current era. (10)

   (b) Write about Electronic Data Interchange in detail. (10)

5. (a) Compare and contrast analog signals with digital signals. (8)

   (b) Narrate the feature and importance of modems with Communication software. (12)

6. Summarize the following concepts :

   (a) multimedia and storage capacity (10)

   (b) optical disks and magnetic tapes used for storage of informations. (10)

7. (a) Give a brief note on various types of database organization. (10)

   (b) Elucidate the role and importance of Management Information System. (10)

8. Explain the following in detail :

   (a) Internet programming with HTML and XML. (12)

   (b) Security issues of information super highway. (8)

DE–3843

Sub. Code
12

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain any four number conversion techniques from one base to another base. (10)

   (b) Describe the number codes and character codes in detail with an example. (10)

2. (a) Write in detail about basic theorems and properties of Boolean algebra. (10)

   (b) Discuss the role of complements in binary arithmetic. (10)

3. (a) State and prove De Morgan's theorem. (10)

   (b) Discuss Quine – Mc Kluskey method with an example. (10)

4. (a) Simplify the Boolean function

   $F(w,x,y,z) = \sum(0,1,\ 2,\ 4,\ 5,\ 6,\ 8,\ 9,\ 12,\ 13, 14)$

   Using Four variable Karnaugh map. (10)

   (b) Explain in detail about multiplexer and demultiplexer with logic diagrams. (10)

5. (a) What are the different ways to represent a negative number? Explain. (10)

   (b) Briefly explain about Half adder and full adder circuits with diagram. (10)

6. (a) Draw the circuits of master – Slave flip – flops and explain. (10)

   (b) Explain the concept of BCD counter with neat diagram. (10)

7. (a) Describe the dynamic random access memory with pin diagram. (10)

   (b) Discuss about magnetic disk memories and magnetic tape memories. (10)

8. (a) Write in detail about Instruction cycle and execution cycle. (10)

   (b) Discuss about he interrupts and interrupt priorities. (10)

DE-3844

Sub. Code
13

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

C AND DATA STRUCTURES

(2005-2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Define

   (i) Identifier

   (ii) Keywords

   Give examples. (10)

   (b) Explain the execution of If statement with examples. (10)

2. (a) What are the data types supported in C? (10)

   (b) Write a program in C to find the factorial value of N (where N is a non negative integer, given as an input). (10)

3. (a) Explain the concept of Recursion. (10)

   (b) Give the importance of Macros. (10)

4. (a) Discuss on any FIVE library functions supported in C. (10)

   (b) Write an program in C to arrange a set of numbers in ascending order. (10)

5. (a) How to pass pointers to function? (10)

   (b) Write a program in C to open a close a data file. (10)

6. (a) How to add an element in a STACK structure? (10)

   (b) How to search an element in a Queue? (10)

7. (a) Compare and contrast a Queue with a stack. (10)

   (b) How to count the number of elements in a Queue. (10)

8. (a) How double linked list performs its job? (10)

   (b) Discuss the methodology of deleting a node in the

      (i) Beginning of a linked list

      (ii) Middle of a linked list

      (iii) End of a linked list. (10)

DE–3845

Sub. Code
14

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about working with multiple windows environment. (10)

   (b) Write down the detailed procedure for how to create a new document in MS-Word. (10)

2. (a) Explain in detail about mail merge concept in MS-Word with an example. (10)

   (b) Write short note on zoom out pages in page orientation. (10)

3. (a) Explain with an example the method of creating charts in MS-Excel. (10)

   (b) What is meant by formatting the text in word? Explain with an example. (10)

4. (a) How to sort the records in Excel database? Explain. (10)

   (b) Describe about the process of adding and removing rows and columns and undoing and redoing in MS-Excel. (10)

5. (a) How to use template in power point? Explain. (10)

   (b) Explain in detail about the various power point views. (10)

6. (a) Explain the steps involved in creating and editing a query. (10)

   (b) Explain in detail about the creation and usage of forms in MS-Access. (10)

7. (a) Explain the method of creating and saving a table in MS-Access. (10)

   (b) Explain the method of Combining Excel data and charts with word documents. (10)

8. Write short notes on the following :

   (a) Page numbering in MS-Word. (8)

   (b) Header and footer in Excel. (6)

   (c) Handouts in Power Point. (6)

DE–7031

Sub. Code
15

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

LAB–I – DATA STRUCTURES USING 'C'

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Break up of marks

Record Note Book : 5
Algorithm(s) /Flow chart(s) : 15
Program(s) : 35
Debugging and Execution : 35
Result(s) /output(s) : 10
Total : 100

Candidate has to answer ONE question chosen by lot system. Each subdivision in a question carries 50 marks.

1. (a) Write a C Program to implement the 'stack' operations such as creation, push, pop, is Full and is Empty using arrays.

   (b) Write a C Program to calculate the factorial of given integer. (60+40)

---------- Cut here ----------

2. (a) Write a C program to implement the 'Queue' operations such as creation, insertion and deletion using arrays.

   (b) Write a C Program to find the given word is a Palindrome or not. (ADA – Palindrome, CPP – not a palindrome) (60+40)

3. (a) Write a C Program to demonstrate string operations/string manipulations using Built-in string functions.

(b) Write a C Program to copy the contents of one file to another file. (50+50)

---------- Cut here ----------

4. (a) Write a C Program to multiply two matrices.

(b) Write a C Program to find the factorial and hence find the value $nCr = \frac{\lfloor n}{\lfloor r \lfloor n-r}$ using function.

(50+50)

---------- Cut here ----------

5. (a) Write a C Program to insert an element and delete an element in a 'Stack'.

(b) Write a C Program to find all the roots of the quadratic equation $ax^2 + bx + c = 0$. (50+50)

---------- Cut here ----------

6. (a) Write a C Program to print the given integer 0 to 9 in towards 0 – ZERO 1 – ONE ..... and 9 – NINE.

(b) Write a C Program to arrange the given 'n' integers in ascending under. (40+60)

7. (a) Write a menu based C program to do the following

getting size of two matrics

getting the values of two matrices

performing addition of two matrices

performing multiplication of two matrices.

(b) Write a C Program to find an element in an array using Linear and Binary search methods. (50+50)

---------- Cut here ----------

8. (a) Write a C Program to evaluate the given mathematical expression using 'stack'.

(b) Write a C Program that concatenate two files : that is append the contents of one file at the end of another and write the result into a third file. (60+40)

DE–7032

Sub. Code
16

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

First Semester

LAB II — MS OFFICE AND INTERNET

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any TWO questions choosing One from Section A and another from Section B.

Each question carries 50 marks.

Breakup of marks
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

SECTION – A

1. Prepare the first page of MCA PRACTICAL RECORD NOTE book with picture insertion and alignment.

---------- Cut here ----------

2. Type lecture notes and provide audio explanation with the help of sound files.

3. Created table of student data that contains Reg no, Name, English, Tamil and Maths marks. Add a new column named TOTAL and find the Mow total for each student. Add two Mows named TOTAL, AVERAGE. Find the total and average values for each subject mark.

---------- Cut here ----------

4. Create a form letter that informs the customer about the date of maturity of a deposit amount in a commercial bank and request the customer for renewal. Mail Merge it with an Access database containing all the customer data. Prepare letters for customers whose due date falls in a specific range of data.

---------- Cut here ----------

5. Create the following worksheet in MS-EXCEL

Item no Name Price Quantity Stock Reorder Purchased Issued on hand Qty level

101 Bolts 2.00 1000 500 300 800 200

(a) Enter all the data items except Qty on hand for 10 items.

(b) Find Qty on hand using the formula.

Stock Qty = Qty on hand + Qty Purchased – Qty issued.

Find total stock value in inventory as a product of total Qty on hand and total price. Display all the items in red color whose Qty on hand is below reorder level.

6. Create a worksheet with the following columns.

Salesman no, Name, Qty, Product name, Sale Amt.

Add three records for 5 different salesman who have carried out sale of different products in different cities.

Find the following :

(a) Qty wise total sales

(b) Salesman wise total sales

(c) Product wise total sales

(d) Grand total sales.

---------- Cut here ----------

7. The following were the observations made in certain experiments for the values y and given the values of x.

x :	1	2	3	4	5	6	7	8	9	10
y :	10	30	45	25	15	28	40	32	15	35

Calculate the following :

(a) Mean, Median and Standard deviation.

(b) Correlation coefficient between x and y.

(c) Draw the bar chart and shading cells.

(d) Draw a chart in Excel and paste it in word.

8. Created data table to create a ready reckoner table for a commercial bank that contains simple interest for Rs. 1,000 with varying period and interest mates. Create the table as shown below.

Ready Reckoner Table for Interest calculation per 1000

INTEREST RATES

PERIOD	3%	4%		11%
1				
2				
.				
.				
.				
10				

---------- Cut here ----------

SECTION – B

9. Create HTML Page to display 10 courses and their highlighted features conducted by a educational institution in a colorful format.

---------- Cut here ----------

10. Create HTML Page that receives employee name, age, street, city, pin data. receive male/female and married/unmarried details through option buttons. Receive hobbies through cheek boxes.

---------- Cut here ----------

11. Create HTML Page that plays video and audio files.

DE–3846

Sub. Code
21

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the elements of Object Oriented Programming. (5)

   (b) What are the necessary steps for compiling a C++ Program? (5)

   (c) How does a main function in C++ differ from main ()in C? (10)

2. (a) Explain briefly about do.. .while loop with sample program segment. (10)

   (b) Write a C++ program to illustrate the concept of switch statement. (10)

3. (a) Discuss in detail about Friend Functions with suitable examples. (10)

   (b) Explain the rules that need to be kept in mind in declaring Virtual Functions. (10)

4. (a) Define class. How does it accomplish data hiding? Explain briefly. (10)

   (b) What is operator overloading? Why is it necessary to overload an operator. (10)

5. (a) Discuss the steps involved in manipulating a file in a C++ Program. (10)

   (b) Describe the various approaches by which the end-of-file condition can be detected. (10)

6. (a) Define Array. Explain briefly about one dimensional array with an example. (10)

   (b) Write a C++ Program to illustrate the concept of Inline Functions. (10)

7. (a) How is a member function of a class defined? Discuss briefly. (10)

   (b) Describe the concept of Recursion with suitable examples. (10)

8. (a) Write a C++ program to illustrate the concept of Multilevel Inheritance. (10)

   (b) Explain briefly about Pointers with sample program segment. (10)

———————

DE–3847

Sub. Code
22

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

SYSTEM ANALYSIS AND DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) With a neat sketch explain the different phases of software development life cycle. (10)

   (b) What are the different types of system? Explain detail. (10)

2. (a) What are the skills required to be a system analyst? Explain. (10)

   (b) Explain interview and questionnaire methods of determining requirements. (10)

3. (a) Describe economical and operational feasibility studies in system analysis. (10)

   (b) Define and explain the procedure for cost benefit analysis.

4. (a) Write down the conventions that govern the construction of DFD. Draw a DFD of student information system. (10)

   (b) Explain the use of decision table. Generate a decision table to determine whether the given year is leap year or not. (10)

5. (a) Explain the logical and physical stages of system design. (10)

   (b) Discuss the methods to design input in online systems. (10)

6. (a) Explain the organization of indexed file with suitable diagrams. (10)

   (b) What is a database? Explain the different types of databases. (10)

7. (a) Describe the steps involved in quality assurance process. (10)

   (b) What is system testing? Explain all the testing strategies in detail. (10)

8. (a) Explain the various conversion methods used in system maintenance. (10)

   (b) Write in detail about the major phases in hardware and software selection. (10)

DE–3848

Sub. Code
23

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) With a neat sketch explain the components of Visual Basic IDE. (10)

   (b) Describe common properties methods and events of VB controls. (10)

2. (a) Discuss the use of scroll bars and timers with examples. (10)

   (b) How does check box differ from Option button? Discuss the properties and events of Option button. (10)

3. (a) Write a VB program to check Whether the given number is odd or even. (10)

   (b) Explain the various looping statements in VB. (10)

4. (a) How to define a function in VB. Explain with an example. (10)

   (b) Write an event procedure in VB to arrange numbers in ascending order. Draw the layout of the User interface also. (10)

5. (a) How will you add and delete controls in a control array? Explain. (10)

   (b) What is a dynamic array? Explain with examples. (10)

6. (a) What is mean by global procedures and Variables? Explain. (10)

   (b) What is MDI form? Explain MDI form creation in VB with example. (10)

7. (a) Explain property procedures with examples. (10)

   (b) What are drive, directory and file list boxes called as file system controls? Discuss. (10)

8. (a) What is OLE? Explain the concepts of OLE with examples. (10)

   (b) Explain Random Access file handling with example. (10)

DE–3849

Sub. Code
24

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Explain the video display generation in detail. (10)

   (b) With an example, explain the ellipse drawing algorithm. (10)

2. (a) With suitable examples, explain 2D basic and composite transformations. (10)

   (b) Write down the Sutherland Hodgman Polygon clipping algorithm. Explain it with an example. (10)

3. Explain the following transformations :

   (a) 3D translation.

   (b) 3D rotation about origin.

   (c) 3D composite transformations (20)

4. Discuss the following

   (a) Window-to-view port transformation. (10)

   (b) 3D clipping. (10)

5. Explain Z-buffer algorithm in detail. (20)

6. Describe the construction of 3D view in detail. (20)

7. Write notes on the following :

   (a) Components of user interface. (10)

   (b) Styles of command language. (10)

8. Explain the concept of information display and feedback in user interface design. (20)

DE–7033

Sub. Code
25

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

COMPUTER LAB III — GRAPHICS AND C++

(2005 to 2010 Calendar Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the candidate by LOT system.

1. (a) Write a procedure and a program in C++ to draw non-linear object circle and three concentric circle.

   (b) Write a program in C++ to draw a 3 – D object and perform animation on it.

---------- Cut here ----------

2. (a) Develop a program to do the following :

   (i) Draw a triangle ABC, translate it to a distance and then rotate it through an angle clockwise.

   (ii) With the same triangle, reverse the operations as in case (i).

   (b) Write a program to draw a ball and perform animation on it.

3. (a) Write a C++ program to draw non-linear object ellipse.

   (b) Write a C++ program to show graphical transformation on three dimension object.

---------- Cut here ----------

4. (a) Write a C++ program to draw a straight line using simple DDA technique.

   (b) Write a C++ program to show graphical transformation of a two dimensional object.

---------- Cut here ----------

5. (a) Create a base class shape, which contains the common elements of circle, triangle and rectangle, from which you derive three classes called ball, rect and tria. Include two member functions set( ) to set coordinates and draw( ) which draws them. Write a main( ) program to exercise this class using graphic – mode graphic functions.

   (b) Write a program to draw a circle. Divide into four quadratic and fill up different color for each quadrant.

---------- Cut here ----------

6. (a) Write a C++ program to draw a straight line using Bresenham's algorithm.

   (b) Write a C++ program to draw non-linear objects arcs and curve.

DE–7034

Sub. Code
26

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2018.

Second Semester

LAB IV — VISUAL BASIC

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each sub division carries 50 marks.

Break-Up	:	
Record	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a VB project to receive a number and to display whether it is a multiple of 5 or not.

   (b) Write a VB project to read the records available in a table and to display them in a grid control.

2. (a) Write a VB project to traverse through the records in database and to display the requested record.

   (b) Write a VB code to create a screen saver that displays 5 pictures with a delay of 1 second each.

---------- Cut here ----------

3. (a) Design a form and write VB code to create a simple calculator and to implement the basic operations.

   (b) Write a VB project to display two lists and to move the item deleted from one into another.

---------- Cut here ----------

4. (a) Write a VB project to receive the employee details in text boxes and to insert it into a table. Also provide facilitates for deletion and updation.

   (b) Write a VB project to create an ActiveX control to receive only numbers with options for setting background and foreground colors.

---------- Cut here ----------

5. (a) Write a VB project to create an editor with few operations like insert, save and open.

   (b) Write VB code to receive a string and to count and display the number of characters.

---------- Cut here ----------

6. (a) Write VB project to read the current usage information like consumer number, name and units consumed from the table and to calculate the tariff (use your own logic).

   (b) Write VB project to move a control by using a timer.

7. (a) Write a VB project to create an activeX control to display a string with dark blue background and yellow background.

   (b) Write a VB project to display nine different images, arranged in 3 rows and 3 columns. The images are covered with buttons and upon checking it, the images should be displayed one by one.

---------- Cut here ----------

8. (a) Write a VB project to create ten buttons with number 1 to 10. By checking the buttons, the corresponding multiplication table is to be displayed.

   (b) Write VB code and design form to display two questions with four options each, to get the answer and to display the results.

DE–3850

Sub. Code
31

DISTANCE EDUCATION

M.C.A./ M.C.A.(Lateral) DEGREE EXAMINATION,
MAY 2018.

Third Semester

DISCRETE MATHEMATICS

(2005-2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. (a) Establish that
$\neg(P \wedge Q) \rightarrow \neg P \vee (\neg P \vee Q))(\neg P \vee Q)$

(b) Obtain the principal conjunctive normal form of the formula. $(\neg P \rightarrow R) \wedge (Q \leftrightarrow P)$.

2. (a) Verify whether $(p \vee q) \vee \neg(p \wedge q)$ is a contradiction or tautology.

(b) Establish the following equivalence :

$(P \vee Q) \wedge (\neg P \wedge (\neg P \wedge Q)) \Leftrightarrow (\neg P \wedge \neg Q)$.

3. (a) If A and B are sets, prove that

$A \Delta B = (A \cup B) - (A \cap B)$

(b) Let A, B and C be sets. R is a relation from A to B and S is a relation from B to C, then $(S \cdot R)^{-1} = R^{-1} \cdot S^{-1}$.

4. (a) Let $f : A \to B$ and $g : B \to C$ be functions. Prove that

(i) if both f and g are one-to-one, $g \circ f$ is also one-to-one.

(ii) if both f and g are onto, $g \circ f$ is also onto.

(b) Let $f : R \to R$ and $g : R \to R$ where R is the set of real numbers be given by $f(x) = x^2 - 2$ and $g(x) = x + 4$. Find $f \circ g$ and $g \circ f$. State whether these functions are injective, surjective and bijective.

5. (a) Let $S = N \times N$, N being the set of positive integers and $*$ be an operation on S given by $(a, b) * (c, d) = (a + c, b + d)$. Show that S is a semigroup. Define :

$f : (S, *) \to (z, +)$ by $f(a, b) = a - b$. Show that f is a homomorphism.

(b) Show that (Z_n, f_n) is an abelian group.

6. (a) Prove that the order of a subgroup of a finite group G divides the order of the group G.

(b) Prove that every subgroup of a cyclic group is cyclic.

7. (a) Show that in a simple digraph $G = \langle V, E \rangle$, every node of the digraph lies in exactly one strong component.

   (b) Set up an algorithm to find the minimum length of path between nodes, when the adjacency matrix of a digraph is given.

8. (a) Prove that a graph is a tree if and only if it is minimally connected.

   (b) Let G be a graph and $e \in E(G)$. Prove that $\tau(G) = \tau(G - e) + \tau(G \cdot e)$.

DE–3851

Sub. Code
32

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about the desirable characteristics of operating system. (10)

   (b) Discuss about the structure of operating system with neat sketch. (10)

2. (a) What is a process? Explain about race conditions in process management. (10)

   (b) Compare and contrast priority scheduling with round robin scheduling. (10)

3. (a) Give the various activities of device controllers in detail. (10)

   (b) Mention and explain the various goals of I/O software. (10)

4. (a) What is a deadlock? Explain the principles and conditions for deadlock. (10)

   (b) What are the functions of Interrupt handlers? Explain. (10)

5. (a) Compare and contrast paging with segmentation. (10)

   (b) Explain about multiprogramming with fixed partitions. (10)

6. (a) Give a detailed note on memory management with bit-maps. (10)

   (b) What is a buddy system? Explain. (10)

7. (a) Explain any two page replacement algorithms with an example. (10)

   (b) What is a file? Explain the different access methods for files. (10)

8. (a) Explain the role of cryptography in security. (10)

   (b) Explain about the various protection mechanisms in file management. (10)

DE–3852

Sub. Code
33

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Third Semester

RDBMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the concept of Database Management System.

   (b) Explain in detail about different User Interfaces.

2. (a) Describe the Relational Model Architecture.

   (b) Explain the comments on Relational Model.

3. (a) Discuss on Data Control Facilities in DBMS.

   (b) Explain the concept of Relation Model.

4. (a) Explain different types of Databases.

   (b) Explain the properties and benefits of Oracle.

5. (a) Explain in detail about Client/Server systems.

   (b) Explain the elements of SQL Language.

6. (a) Explain the DML commands in Oracle.

   (b) How to work with NULL values in Oracle? Explain.

7. (a) Explain unions and multiple part queries in Oracle.

   (b) Explain the following :

      (i) Create Table

      (ii) Dropping a table.

8. (a) Discuss on Indexes in Oracle.

   (b) Explain about Embedded SQL statements with an example.

DE–3853

Sub. Code
34

DISTANCE EDUCATION

M.C.A. / M.C.A (Lateral) DEGREE EXAMINATION, MAY 2018.

Third Semester

UNIX AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain briefly about different types of shells available in Unix OS. (10)

   (b) What are the different types of files in UNIX operating system? Explain. (10)

2. (a) Discuss on command arguments and parameters in Unix. (10)

   (b) Explain the conditional expressions in UNIX with suitable examples. (10)

3. (a) Elucidate about While and Until loops with examples. (10)

   (b) Explain how to track file changes in UNIX. (10)

4. (a) Discuss in detail about UNIX system calls. (10)

   (b) What is standard input and output? Explain about Shell input/output redirections. (10)

5. (a) Explain the need of Tbl and eqn preprocessors. (10)

   (b) Discuss the control flow and relational operators. (10)

6. (a) What is the difference between internal command and external command? Explain the general format of command with examples. (10)

   (b) Explain the three modes of the Vi-Editor. (10)

7. (a) Write a detailed note on Shell variables. (10)

   (b) How do you execute a command in the shell? Explain. (10)

8. (a) Summarize about Zap and Pick commands in UNIX with examples. (10)

   (b) Explain in detail about Signal and Interrupts. (10)

DE–3854

Sub. Code
35

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the process of communication.

   (b) What are the various reading styles?

2. (a) How is an agenda written?

   (b) What are the commandments of Communication?

3. (a) Discuss the art of listening.

   (b) Explain the process of brain storming.

4. (a) What are the interview techniques?

   (b) How is a mocking interview conducted?

5. (a) Comment on the art of guiding.

   (b) How is personality tested through group discussion?

6. (a) How is body language important in communication?

   (b) Explain lateral thinking.

7. (a) What are the various negotiation techniques?

   (b) What are the goals of conducting a meeting?

8. (a) Explain downward communication.

   (b) What is the role of union in communication?

DE–7035

Sub. Code
36

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Third Semester

LAB V — RDBMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

ONE question is to be given by the Examiner to each candidate by LOT system.

Break-up marks

Record : 5
Algorithm : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Write a program in ORACLE for hospital billing system with the following fields:

Patient number, name, age, doctor name, patient type, consulting charge, test charges and total fee.

Prepare reports for the following conditions :

(i) The patients who belong to In-patient category.

(ii) The patients undergone for test.

(iii) List of patients is the age between 20 to 28.

(b) A daily sales file contains records with the following fields :

Department number, date, item description, sale price for each item, quantity, cost of each item.

Write a program using ORACLE to list all the input data and compute total amount of sales and profit for each sale. Print total sales and profit. Show the output report in the following format :

Department number

Item description

Sales price

Cost

Profit

---------- Cut here ----------

2. (a) Write an SQL program for the Canteen maintenance, with two tables namely MENU and BILL.

The MENU table contains the following fields:

ITEM, UNIT-RATE.

The following items are available in the Canteen:

TEA, COFFEE AND COOL-DRINKS

The BILL table contains the following fields:

EMP NO, EMP NAME, DATE OF ISSUE, ITEM1, NUMBER OF TOKENS ISSUED FOR ITEM1, RATE1, ITEM2, NUMBER OF TOKENS ISSUED FOR ITEM2, RATE2, ITEM3, NUMBER OF TOKENS ISSUED FOR ITEM3, RATE3, TOTAL AMOUNT.

(b) Write a PLSQL program to prepare a mark statement of a student using STUDENT Table and MARK Table.

3. (a) An airline reservation database contains the reservation table and personal table. Assume your own database.

Write a program to prepare the following reports.

(i) Total number of male passengers in a particular flight.

(ii) List out the female passages in the age between 18 to 25 for all flights.

(b) A salary statement contains Name, basic pay, allowances, deductions, Gross pay and Net pay.

Write a program in SQL to prepare salary report for 5 employees. Assume your own data base.

---------- Cut here ----------

4. (a) A hospital maintains blood doners information in a file. Assume your own database. Write a program using oracle to print out the number, name and address of the doners having AB blood group and in the age group of 16 to 25.

(b) Write a SQL program to prepare a bill for hotel with two tables namely MENU and BILL. Assume your own database.

DE–7036

Sub. Code
37

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2018.

Third Semester

LAB : VI — ALGORITHM AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Break-up marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

ONE question is to be given by the examiner to each candidable by lot system.

1. (a) Write an algorithm to implement the concept of straight insertion sorting.

(b) Write a shell program to print the access rights list of a given file.

2. (a) Write down the procedures for sequential and binary search methods.

   (b) Write a shell program to check for the existence of a given pattern in a file using arguments.

---------- Cut here ----------

3. (a) Write an algorithm for arranging the numbers in increasing order using partition (Quick) sorting.

   (b) Write a shell program to implement a calculator with the basic trigonometric functions.

---------- Cut here ----------

4. (a) Write an algorithm for tree (heap) sorting.

   (b) Write a menu driven program to manipulate the files.

---------- Cut here ----------

5. (a) Write an algorithm to find the least cost tour made a sales person using Branch and Bound method.

   (b) Write a shell program to find the sum of the following series:

$$\frac{1}{2!}+\frac{2}{3!}+\frac{3}{4!}+\ldots+\frac{n}{(n+1)!}.$$

6. (a) Write an algorithm for solving the Towers of Hanoi problem having 'N' number of disks.

   (b) Write a shell program to prepare the payroll for an organization having different classes of employees.

---------- Cut here ----------

7. (a) Write down the procedure for straight exchange sorting.

   (b) A famous medical research centre maintains the profile of all medical practitioner who are working in it. Write a shell program to print the list of inmates according to their profession alphabetically. Also, write a program segment to print the personnel working in the department of Cardiology.

---------- Cut here ----------

8. (a) Write an algorithm to print the traversals made by a Knight on a chess board.

   (b) Write a shell program to send a mail to a group of users.

DE–3855

Sub. Code
41

DISTANCE EDUCATION

M.C.A. / M.C.A (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions only.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write short notes on:

   (i) Unicasting

   (ii) Bluetooth

   (iii) DSL

   (iv) Bandwidth. (2+2+2+2)

   (b) Explain the functions and responsibilities of the layers of OSI network model. (12)

2. (a) Explain any two guided and one unguided transmission media in detail. (10)

   (b) How does broadband ISDN help us in communication networks? (10)

3. (a) With an example, write and explain the algorithm for checksum calculation. (10)

   (b) Explain the different versions of ALOHA protocol. (10)

4. (a) How does Go-sack-N facilitate us in dealing the errors in pipelining? Explain. (10)

   (b) Give an account on High Level Data link Control bit-oriented protocol. (10)

5. (a) Explain the concept of Distance Vector Routing in detail. (10)

   (b) Explain the role of IP in communication. (10)

6. How does transport layer ensure the source-to-destination delivery of entire message? (20)

7. (a) Write notes on : (i) DNS and (ii) RSA. (6+7)

   (b) Write down the rules for obtaining optimum mileage in network performance. (7)

8. (a) Write an essay on Multimedia. (12)

   (b) Write short dotes on :

      (i) Voice over IP and

      (ii) Video Compression. (4+4)

DE–3856

Sub. Code
42

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) "Accounting is the language of business' – Elucidate.

   (b) Describe the advantages and limitations of accounting.

2. (a) Write down the advantages and disadvantages of budgetary and explain it in detail.

   (b) What do you understand by perpectual inventory system? Explain.

3. From the given details prepare trading and P and L a/c and balance sheet of Ms. Lakshmi and firm.

Debit balance	Rs.	Credit balance	Rs.
Sundry debtors	92,000	Lakshmi's capital	70,000
P and M	20,000	Purchase return	2,600

Debit balance	Rs.	Credit balance	Rs.
Interest	430	Sales	2,50,000
Rent	5,600	Sundry creditors	60,000
Conveyance charges	1,320	Bank OD	20,000
Wages	7,000		
Sales return	5,400		
Purchase	1,50,000		
Lakshmi drawing	22,000		
Trade expenses	1,350		
Opening stock	60,000		
Salary	11,200		
Advertising	840		
Discount	600		
Bad debts	800		
Business premius	12,000		
Furniture	10,000		
Cash in hand	2,060		
	4,02,600		4,02,600

Adjustments :

(a) Stock on 31.12.2015 is Rs. 90,000.

(b) Provide depreciation on premises @ 2.5%, P and M @ 7.5% and furniture @ 10%.

(c) Write off ₹ 800 as further bad debts.

(d) Provide for doubtful debts @ 5% on sundry debtors.

(e) Outstanding rent was ₹ 500 and at sundry wages ₹ 400.

(d) Prepaid rent ₹ 300 and prepaid salary ₹ 700.

4. You are given the following informations :

	Rs.
Cash	18,000
Debtors	1,42,000
Closing stock	1,80,000
B/P	27,000
Creditors	50,000
Outstanding expenses	15,000
Tax payable	75,000

Calculate (a) current ratio (b) liquid ratio (c) absolute liquidity ratio.

5. From the comparative B/S of Rajlakshmi Ltd as on 30th June 2015 and 2016. Prepare funds flow statement and statement of charges of working capital.

Balance sheet.

Liabilities	30.6.2015 Rs.	30.6.2016 Rs.	Assets	30.6.2015 Rs.	30.6.2016 Rs.
Share capital	1,80,000	2,00,000	Goodwill	24,000	20,000
Reserve fund	28,000	36,000	Buildings	80,000	72,000
P and L a/c	39,000	24,000	Machinery	74,000	72,000
Trade creditors	16,000	10,800	Investment	20,000	22,000

Liabilities	30.6.2015 Rs.	30.6.2016 Rs.	Assets	30.6.2015 Rs.	30.6.2016 Rs.
OD	12,400	2,600	Inventories	60,000	50,800
Provision for tax	32,000	34,000	Debtors	40,000	44,400
Provision for doubtful debts	3,800	4,200	Cash	13,200	30,400
	3,11,200	3,11,600		3,11,200	3,11,600

Additional information :

(a) Depreciation on machinery Rs. 10,000 and building Rs. 8,000.

(b) Investment sold during the year Rs. 3,000.

(c) Tax paid during the year Rs. 30,000.

(d) Rs. 15,000 interest dividend paid in Jan 2016.

6. Prepare shares ledger under FIFO method for the given information :

1 Jan 2016	Balance	800 units @ Rs. 9 each
3 Jan 2016	Purchase	500 units @ Rs. 9.50 each
5 Jan 2016	Issued	900 units
8 Jan 2016	Purchase	900 units @ Rs. 9.30 each
12 Jan 2016	Issued	300 units
15 Jan 2016	Issued	400 units
17 Jan 2016	Issued	200 units
20 Jan 2016	Purchase	500 units @ 9.5 paise each
21 Jan 2016	Issued	300 units

7. Vasanth trading company presents the following information. Calculate the

(a) PV ratio

(b) BEP and margin safety

	Rs.
Sales	2,00,000
Variable cost	1,20,000
Fixed cost	50,000
Net profit	30,000

8. Prepare a flexible budget on the basis of the following data ascertain overhead rates @ 50%, 60% and 70% capacity.

Variable overheads	@ 60% capacity
	Rs.
Indirect material	6,000
Indirect labour	18,000
Semi variable overheads	
Electricity (40% fixed)	30,000
Repairs (20% variable)	3,000

Fixed overheads	
Depreciation	16,500
Insurance	4,500
Salary	15,000
Total overheads	93,000

Estimated direct labour hours : 1,86,000.

DE–3857

Sub. Code
43

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Use simplex method to solve the following L.P.P.

Maximize $z = x_1 + 2x_2$

Subject to the constraints :

$-x_1 + 2x_2 \leq 8$

$x_1 + 2x_2 \leq 12$

$x_1 - 2x_2 \leq 3$

$x_1 \geq 0$ and $x_2 \geq 0$. (20)

2. Consider the following transportation problem.

Factory	Godowns						Stock available
	1	2	3	4	5	6	
A	7	5	7	7	5	3	60
B	9	11	6	11	–	5	20
C	11	10	9	6	9	12	90
D	9	10	9	6	9	12	50
Demand	60	20	40	20	–	40	

It is not possible to transport any quantity from factory B to Godwon 5.

Determine :

(a) Initial solution by Vogel's approximation method.

(b) Optimum basic feasible solution.

(c) Is the optimum solution unique? If not, find the alternative optimum basic feasible solution. (20)

3. (a) Write down the algorithm for Hungarian assignment method. (10)

(b) Arrivals at a telephone booth are considered to be Poisson, with an average time of 10 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponential, with mean 3 minutes.

(i) What is the probability that a person arriving at the booth will have to wait?

(ii) The telephone department will install a second booth when convinced that an arrival would expect waiting for at least 3 minutes for phone. By how much should the flow of arrivals be increased in order to justify a second booth?

(iii) Find the average number of units in the system.

(iv) Estimate the fraction of a day that the phone will be un use.

(v) What is the probability that it will take him more than 10 minutes altogether to wait for phone and complete his call? (10)

4. (a) A super market has two girls running up sales at the counters. If the service time for each customer is exponential with mean 4 minutes, and if people arrive in a Poisson fashion at the rate of 10 an hour,

(i) What is the probability of having to wait for service?

(ii) What is expected percentage of idle time for each girl?

(iii) If a customer has to wait, what is the expected length of his waiting time? (12)

(b) Write down the simulation methodology. (8)

5. (a) Find the optimum order quantity for a product for which the price breaks are as follows :

Quantity	Unit cost (Rs.)
$0 \le Q_1 < 800$	Re. 1.00
$800 \le Q_2$	Re. 0.98

The yearly demand for the product is 1600 units per year, cost of placing an order is Rs. 5, the cost of storage is 10% per year. (12)

(b) Narrate the basic components and rules for network construction. (8)

6. The following table gives the activities in a construction project and other relevant information.

Activity $i-j$	Normal time (days)	Crash time (days)	Normal cost (Rs.)	Crash cost (Rs.)
1-2	20	17	600	720
1-3	25	25	200	200
2-3	10	8	300	440
2-4	12	6	400	700
3-4	5	2	300	420
4-5	10	5	300	600
4-6	5	3	600	900
5-7	10	5	500	800
6-7	8	3	400	700

(a) Draw the activity network of the project.

(b) Find the total float and free float for each activity.

(c) Using the above information crash the activity step by step until all paths are critical. (20)

7. (a) A barber shop has two barbers and three chairs for waiting customers. Assume that customers arrive in a Poisson fashion at a rate of 5 per hour and that each barber services customers according to an exponential distribution with mean of 15 minutes. Further, if a customer arrives and there are no empty chairs in the shops he will leave. Find the steady state probabilities. What is the probability that the shop is empty? What is the expected number of customers in the shop? (12)

(b) Narrate the advantages and limitations of simulation. (8)

8. (a) A firm is considering replacement of a machine, whose cost price is Rs. 12,200 and the scrap value is Rs. 200. The running (maintenance and operating) costs in Rs. are found from experience to be as follows :

Year :	1	2	3	4	5	6	7	8
Running cost :	200	500	800	1200	1800	2500	3200	4000

When should the machine be replaced? (10)

(b) Let the value of money be assumed to be 10% per year and suppose that machine A is replaced after every 3 years where as machine B is replaced after every six years. The yearly costs of both the machines are given below :

Year :	1	2	3	4	5	6
Machine A :	1000	200	400	1000	200	400
Machine B :	1700	100	200	300	400	500

Determine which machine should be purchased. (10)

DE–3858

Sub. Code
44

DISTANCE EDUCATION

M.C.A. / M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write Short Notes on: The Evolution of the Object Model. (10)

   (b) Discuss about Applying Object Model. (10)

2. (a) What are the Relationships among Objects? Explain with examples. (10)

   (b) Write Short Notes on: The Interplay of Classes and Objects. (10)

3. (a) Discuss on Building Quality Classes and Objects. (10)

   (b) Discuss the importance of Proper Classification. (10)

4. (a) How to identifying Proper Classes and Objects? Explain. (10)

   (b) Write Short Notes on: Meta Model. (10)

5. (a) Discuss about Inception and Elaboration. (10)

(b) Write Short Notes on: Patterns. (10)

6. (a) Discuss briefly on: Use Cases. (10)

(b) Write Short Notes on: UML Class Diagram. (10)

7. (a) Discuss about Static and Dynamic Models. (10)

(b) Write about UML Interaction Diagrams. (10)

8. (a) Write Short Notes on : UML Activity Diagram. (10)

(b) Explain about UML and Programming. (10)

DE–3859

Sub. Code
45

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the features of JAVA in Internet Programming. (10)

   (b) Discuss about Security and Encryption standards. (10)

2. (a) Describe the features of VB. Script. (10)

   (b) Discuss the functionality of CGI. (10)

3. (a) Describe the HTML tags that are used for formatting and frame design. (10)

   (b) Explain nested list in HTML. (10)

4. (a) Write in detail about the Microsoft Internet explorer. (10)

   (b) Explain shokwave and lingo. (10)

5. (a) How to create a custom integrated application with multiple protocols? (10)

   (b) How do you pull web information? (10)

6. (a) How to create an activex control to activate a web page? (10)

   (b) Discuss about VDO live Technology. (10)

7. (a) Write in detail about animation in Internet Programming. (10)

   (b) Discuss about Images and Hyperlinks. (10)

8. (a) How do you add graphics to web pages? (10)

   (b) Write short notes on site and page Design Concepts. (10)

DE–7037

Sub. Code
46

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

LAB VII — INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a VB Script to do the following

   (i) Check the given Password

   (ii) Change the existing Password

   (b) Design and publish a web page for a college.

---------- Cut here ----------

2. (a) Write a Java Script to create an Order Form to select the house articles.

   (b) Develop a picture gallery having at least 2 pages. Each of them is having several pictures with suitable information.

3. (a) Write a Swing program to create the Tabbed panels.

   (b) Create a web page in the format of front page of a news paper using Text link. Align the text with colors.

---------- Cut here ----------

4. (a) Write a Java program to create a color as the background.

   (b) Develop a single page advertisement for a shop to be opened shortly.

---------- Cut here ----------

5. (a) Write a Java Script to create a color Palet and display the text in the color chosen from the Palet with proper background color.

   (b) Design and publish a web page for a college.

---------- Cut here ----------

6. (a) Write a VB Script to create a calendar for given Month and Year.

   (b) Create a web page in the format of front page of a news paper using Text link. Align the text with colors.

---------- Cut here ----------

7. (a) Write a Java program using Applet to create the frames and its controls.

   (b) Develop a single page advertisement for a shop to be opened shortly.

8. (a) Write a Java program using Applet to display the dialogue menu in applet.

(b) Develop a web page for job recruitment agency in an IT industry.

DE–7038

Sub. Code
47

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fourth Semester

LAB VIII — NETWORK LAB

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write a Java program to add two numbers using RMI client, server technique.

   (b) Write a Java program to read a file from the remote system.

---------- Cut here ----------

2. (a) Write a Java program to find factorial of given number using RMI client, server technique.

   (b) Write a Java program to send messages from one system to another system.

---------- Cut here ----------

3. (a) Write a Java program to find factorial of given number using FACTORIAL INTERFACE technique.

   (b) Write a Java program to setup Time Server and Time Client.

4. (a) Write a Java program to setup a chat server and chat client.

   (b) Write a Java program to ENCRYPT given word.

---------- Cut here ----------

5. (a) Write a Java program to setup echo server and echo client.

   (b) Write a Java program to display the users and their IP addresses logged in the network.

DE–3860

Sub. Code
51

DISTANCE EDUCATION

M.C.A. / M.C.A (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain briefly about the elements of digital geometry. (10)

   (b) Discuss in detail about perspective projection technique. (10)

2. (a) Write in detail about sampling and quantization with diagrams. (10)

   (b) Write brief note on low-pass filtering. (10)

3. (a) Write a detailed note on restoration by maximum posterior estimation. (10)

   (b) Discuss on the following.

      (i) Image sharpening (5)

      (ii) Image averaging. (5)

4. (a) Describe in detail about geometric transformation with suitable example. (10)

(b) What is the necessity for image compression? Discuss Huffman coding with suitable examples. (10)

5. (a) Explain the various steps involved in processing color images. (10)

(b) Discuss the advantages and disadvantages of lossy compression and loss-less compression. (10)

6. (a) Distinguish ranging from multi level thresholding and local thresholding. (10)

(b) Describe about line detection and corner detection. (10)

7. (a) Explain the tree search method for feature extraction with suitable examples. (10)

(b) Write in detail about stereo imaging algorithms. (10)

8. (a) Write short notes on the following.

(i) Tree search. (5)

(ii) Graph matching. (5)

(b) Explain about fuzzy mathematical recognition in detail. (10)

———————

DE–3861

Sub. Code
52

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the various activities involved in developing and maintaining a software product. (10)

   (b) Explain any ten factors which influence the productivity and quality. (10)

2. (a) What is meant by software process model? Explain any two process models in detail. (2+6+6)

   (b) Write a note on Process Metrics. (6)

3. (a) Explain the steps involved in planning for a software project. (15)

   (b) With the structure and communications paths, explain the features of democratic team. (5)

4. (a) How do we estimate the software cost through COCOMO? (8)

   (b) Explain the different types of project structure in detail. (12)

5. (a) Give an account on Risk Management. (12)

   (b) List down the activities performed by the SQA team. (8)

6. Write short notes on : $(4 \times 5 = 20)$

   (a) Software Configuration Management

   (b) Characteristics of a SRS.

   (c) Walkthroughs

   (d) FURPS

7. (a) Explain the golden rules for designing an efficient interface. (10)

   (b) What is a module? What are its characteristics? What are the measures used to design an effective module? (10)

8. (a) Explain the basis path testing in detail. (10)

   (b) Draw a structured flowchart and flow graph to find the biggest of three numbers. Also, determine the numbers of test cases required to verify and validate the above mentioned task through cyclomatic complexity. (10)

———————

DE–3862

Sub. Code
53

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

VISUAL PROGRAMMING – II

(2005–2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Mention the salient features of visual C++. (10)

   (b) Discuss the steps involved in creating application using application wizard. (10)

2. (a) Explain the technique for displaying windows messages. (10)

   (b) Discuss the following

      (i) Document object

      (ii) View object. (10)

3. (a) What is the need for maximze and minimize buttons? Describe the process involved in it. (10)

   (b) How to add menus to the programs. (10)

4. (a) Give the importance of message box function. (10)

   (b) Write a program to create your own pen to draw a square and a rectangle. (10)

5. (a) Give the purpose of combo box. (10)

   (b) What is a Notepad? How to create it? (10)

6. (a) How member variables can be used to update data? (10)

   (b) Explain the technique of using VBX controls in code. (10)

7. (a) Give the importance of multiple document interface programs. (10)

   (b) How to add scroll bars? (10)

8. (a) How to programe OLE in VC++? (10)

   (b) Mention the salient features of ODBC classes in VC++. (10)

DE-3863

Sub. Code
54

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain about the role of lexical analyzer. (10)

   (b) Briefly explain about the structure of a compiler. (10)

2. (a) What are the difficulties with generalized top down parsing? (10)

   (b) State the actions of a shift-reduce parser on the input string $id_1 + id_2 * id_3$ according to the grammar

   E->E + E

   E->E*E

   E-> (E)

   E->id. (10)

3. (a) Construct LALR parsing table for the grammar

   S->CC

   C->cC | d. (10)

   (b) Give a brief note on automatic parser generator. (10)

4. (a) How to implement syntax-directed translation? Explain. (10)

   (b) Write the translation scheme for generating 3-address code for the CASE statement. (10)

5. (a) Generate intermediate code for the following code segment along with the required syntax directed translation scheme

```
while (i<10)
if (i%2==0)
evensum=evensum +i;
else
oddsum=oddsum +i;
```

(10)

   (b) Write short note on postflx translations. (10)

6. Describe in detail about the contents and data structures of a symbol table. (20)

7. (a) Write down the principle source of optimization. (10)

   (b) Discuss on lexical-phase errors briefly. (10)

8. (a) Discuss the importance of peephole optimization. (10)

   (b) Explain the use of DAG in analyzing basic blocks with an example. (10)

———————

DE–3864

Sub. Code
55

DISTANCE EDUCATION

M.C.A. / M.C.A (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss in brief about Data mining in marketing. (10)

   (b) Explain how information acts as a production factor. (10)

2. Discuss the various stages of Knowledge Discovery Process with an example. (20)

3. Describe the process architecture of data warehouse with suitable illustration. (20)

4. (a) What is a Parallel Query? How does parallel query work? (10)

   (b) Describe about backup and recover operations in data warehouse. (10)

5. Explain in detail about Capacity planning for a data warehouse. (20)

6. (a) Give a detailed note on Concept learning. (10)

   (b) Describe about OLAP tools. (10)

7. (a) Discuss on association rules with an appropriate example. (10)

   (b) Explain in detail about neural network and its applications. (10)

8. (a) Elucidate how to design fact tables and dimension tables. (10)

   (b) Discuss the day-to-day operations of the data warehouse. (10)

DE–7039

Sub. Code
56

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

LAB–IX : VISUAL C++

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by LOT system.

1. (a) Write Visual C++ win 32 application program using MFC that allows the user to draw pictures with the help of mouse as a free hand drawing tool.

   (b) Write Visual C++ win 32 application program using MFC that creates two push buttons OK and CANCEL on the client area. Buttons should respond to user click over them and display appropriate message.

---------- Cut here ----------

2. (a) Write Visual C++ win 32 application program using MFC that displays a message "Hello Good Morning!" Wherever the user clicks the mouse button on the client area.

   (b) Write Visual C++ win 32 application program using MFC that displays current mouse coordinates in status bar.

3. (a) Write Visual C++ win 32 application program using MFC that creates a list box and displays name of the states in India.

(b) Write Visual C++ 32 application program using MFC that displays the status of ALT, CTRL, SHIFT, NUMLOCK and SCROLL LOCK keys.

---------- Cut here ----------

4. (a) Write Visual C++ win32 application program using MFC that displays line, rectangle, rounded rectangle, ellipse and polygon filled with colors.

(b) Write Visual C++ win 32 application program using MFC that displays a menu. Choose the menu items using keyboard accelerator keys and display appropriate messages for the selected command in message box.

---------- Cut here ----------

5. (a) Write Visual C++ win 32 application program using MFC that create a new font.

(b) Write Visual C++ win 32 application program using MFC that fills the background of the client area with a bitmap.

DE–7040

Sub. Code
57

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Fifth Semester

LAB X — COMPILER DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give One question to each candidate by LOT system.

Write program in C language for any ONE of the following :

1. Write a C program to recognize identifiers in a source program.

---------- Cut here ----------

2. Write a C program for drawing transistion diagram for given regular expression.

---------- Cut here ----------

3. Write a program in C to check if a given grammar is an operator grammar or not.

---------- Cut here ----------

4. Write a program in C, to decide whether a given string is derivable from a given context free grammar.

5. Write a C program to construct SLR parsing table for a context free grammar.

---------- Cut here ----------

6. Write a C program to convert an infix expression into equivalent postfix firm.

---------- Cut here ----------

7. Write a C program to identify the floating constants in a source program.

---------- Cut here ----------

8. Write a C program to recognize keywords in a source program.

---------- Cut here ----------

9. Write a C program to translate an arithmetic expression into triples and indirect triples.

---------- Cut here ----------

10. Write a C program to construct predictive parser for the following grammar.

$E \rightarrow E + T / T, T \rightarrow T * F / F, F \rightarrow (E) / id$.

DE–3865

Sub. Code
61

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Describe the activities covered by software project management with suitable illustrations. (20)

2. Explain briefly about the three approaches for identifying the activities or tasks that make up a project with appropriate examples. (20)

3. Discuss in detail about the main objectives of risk management. (20)

4. Describe the salient features of software configuration management. Give a suitable example for software configuration management. (20)

5. Explain in detail about the components of PRINCE 2 project management. (20)

6. Enumerate the problems that are encountered with software projects with appropriate examples. (20)

7. (a) Define resource. Explain briefly about its categories. (10)

   (b) Describe a set of change control procedures that would be appropriate for software project. (10)

8. Identify the key aspects in which modern software project management practices differ from those of traditional software project management practices. Justify your answer. (20)

DE–3866

Sub. Code
62

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Define : Multimedia. Write its need and benefits. (10)

   (b) Write a note on multimedia platforms. (10)

2. (a) Discuss about commercial tools of multimedia. (10)

   (b) Brief a note on non-temporal media types. (10)

3. (a) Explain about extended images and digital ink. (10)

   (b) Describe about redundancy in compression technique. (10)

4. (a) Illustrate Image Compression concept. (10)

   (b) Discuss about objects and classes. (10)

5. (a) Write a note on media classes. (10)

   (b) Describe about transform and format classes. (10)

6. (a) Brief a note on components classes. (10)

   (b) Discuss about the CD family. (10)

7. (a) Explain about media organization in detail. (10)

   (b) Brief a note on Architecture and operations. (10)

8. (a) Explain about multimedia on network. (10)

   (b) Multimedia and single user – Discuss. (10)

DE–3867

Sub. Code
63

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain briefly about the architecture of neural networks. (10)

   (b) Discuss about the benefits of ANN. (10)

2. (a) Explain about the various learning tasks. (10)

   (b) Write brief note on competitive learning. (10)

3. (a) Briefly explain the multilayer training algorithms. (15)

   (b) What are the limitations of back propagation networks? (5)

4. (a) Explain the relation between perceptron and bayes classifier in gaussian environment. (10)

   (b) Explain the various applications of Hop field network. (10)

5. (a) Define perceptrons. Explain its various algorithms. (10)

   (b) Discuss in detail about associative memory. (10)

6. (a) Explain briefly about the energy function for bidirectional associative memory. (10)

   (b) Discuss in detail about least mean-square algorithm. (10)

7. (a) Compare cognitron and neocognitron. (10)

   (b) Discuss briefly about the overview of adaptive resonance theory. (10)

8. (a) Explain briefly about the holographic correlators. (10)

   (b) Describe in detail about the structure of optical neural network. (10)

DE–3868

Sub. Code
64

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

C-SHARP (C#)

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Mention the salient features of Java.

   (b) Discuss on microsoft net framework.

2. (a) Discuss on the components of net.

   (b) How error handling is done in C#.

3. (a) Mention the salient features of C#.

   (b) Discuss on JITers.

4. (a) Explain about common language specification.

   (b) Discuss on virtual execution system.

5. (a) Brief a note on C# data types with example.

   (b) How to choose the editor for C#?

6. (a) Discuss about constructors and destructor methods with example.

   (b) Explain the class properties with examples.

7. (a) Describe about Exception Handling.

   (b) How to work with Name spacer?

8. (a) Discuss about conditional compilation.

   (b) How documentation and comments are done in XML?

DE–7041

Sub. Code
65

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2018.

Sixth Semester

LAB – XI (MULTI MEDIA APPLICATIONS)

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give one question to each candidate by Lot system.

1. (a) Write a C++ program to find the edges of the following image using sobel edge detection procedure.

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 2 & 2 & 0 & 3 & 3 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 & 4 & 2 & 2 \\ 0 & 2 & 0 & 0 & 2 & 4 & 3 & 2 & 3 & 3 \\ 0 & 1 & 3 & 3 & 4 & 4 & 3 & 3 & 3 & 3 \\ 0 & 1 & 0 & 4 & 3 & 3 & 2 & 4 & 3 & 3 \\ 1 & 0 & 0 & 1 & 2 & 3 & 4 & 2 & 3 & 4 \end{bmatrix}$$

(b) Prepare a multimedia presentation shows the features of a new model mobile phone.

---------- Cut here ----------

2. (a) Write a C++ program for two dimensional image animation using computer graphics.

(b) Design a wedding day greeting page including various graphic features in flash.

3. (a) Write a C++ program to find the best grey level for splitting the following image :

$$\begin{bmatrix} 4 & 5 & 1 & 7 & 4 & 2 & 6 & 3 & 4 & 2 & 7 & 4 & 6 & 7 \\ 3 & 7 & 8 & 2 & 7 & 6 & 5 & 4 & 3 & 3 & 7 & 2 & 4 & 3 \\ 4 & 5 & 9 & 9 & 9 & 7 & 8 & 7 & 6 & 5 & 5 & 4 & 1 & 2 \\ 0 & 0 & 0 & 3 & 4 & 2 & 2 & 3 & 1 & 1 & 4 & 6 & 6 & 5 \\ 7 & 3 & 2 & 4 & 5 & 1 & 6 & 2 & 3 & 4 & 5 & 6 & 7 & 9 \end{bmatrix}$$

(b) Design a sales advertisement for a product using Flash.

---------- Cut here ----------

4. (a) Write a C++ program for grey level transformation techniques.

(i) Thresholding

(ii) Bunching.

(b) Do the image animation using Flash.

---------- Cut here ----------

5. (a) Write a C++ program for setting up a pattern for multi–grey–level images or bilevel display using limbs algorithm with the following 2×2 matrix : $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}$ Also print the dot patterns produced on a 16 grey–level system for a pixel.

(b) Design an animated Cartoon picture.

---------- Cut here ----------

6. (a) Write a C++ program to compress the image using contour coding technique. Assume your own image matrix.

(b) Prepare a multimedia presentation showing the features of the courses offered in a college.